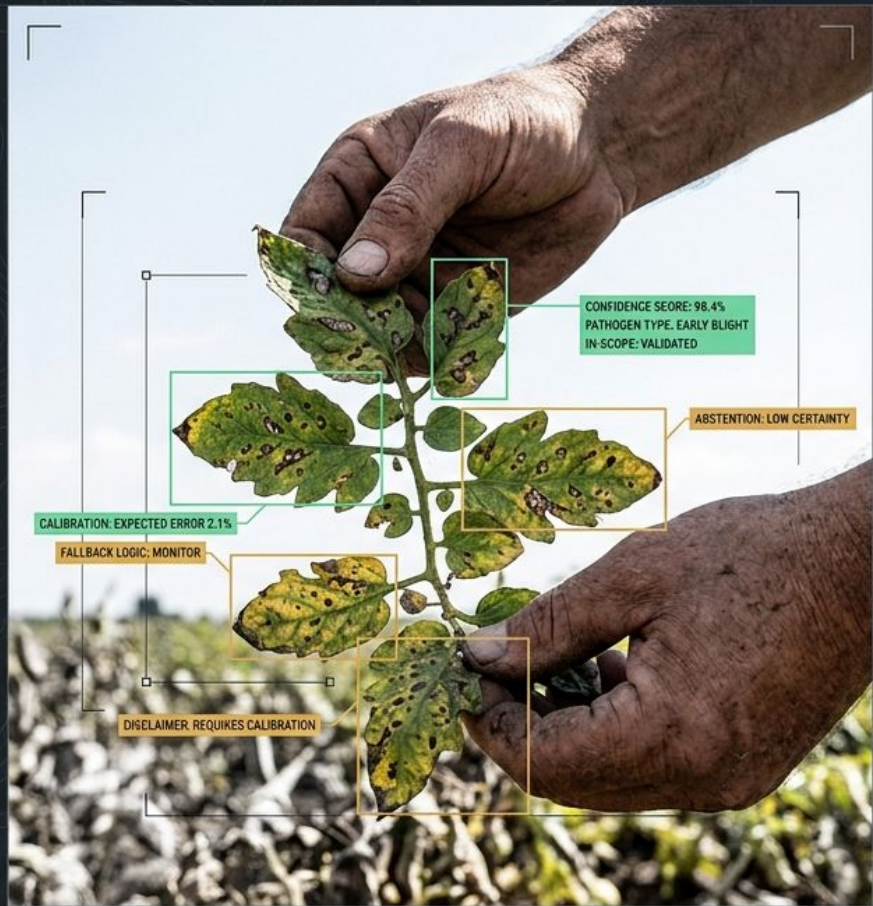




ROOTED IN TRUTH

A Field-Robust AI Crop Advisory System

Milestone 1 Technical Presentation

THE TWO BIGGEST TRUST FAILURES IN AI AGRICULTURE



THE PROMISE:
FIELD-ROBUST
VISION

THE FAILURE:

Models hit **99% in the lab**, but **collapse to ~41% in real fields**.

THE PROOF:

F1 gap quantified on real-world datasets (PlantDoc, Cassava).



THE PROMISE:
CONFIDENCE-AWARE
ANSWERS

THE FAILURE:

Systems **blindly guess even when clueless**.
Telling a farmer "spray Mancozeb" at 38% confidence is **harmful**.

THE PROOF:

Expected **Calibration Error (ECE)** and **abstention correctness thresholds**.



THE PROMISE:
FAITHFULNESS-
GROUNDED ADVICE

THE FAILURE:

LLMs **hallucinate dangerous pesticide doses**.

THE PROOF:

Every claim **traces to cited sources**, measured by RAGAS metrics.

EVERY CLAIM MADE BY THIS SYSTEM IS BACKED BY A MEASURABLE NUMBER.
WE DO NOT RELY ON FLUENCY TO FAKE ACCURACY.

The Lab-to-Field Cliff

The Lab (PlantVillage)

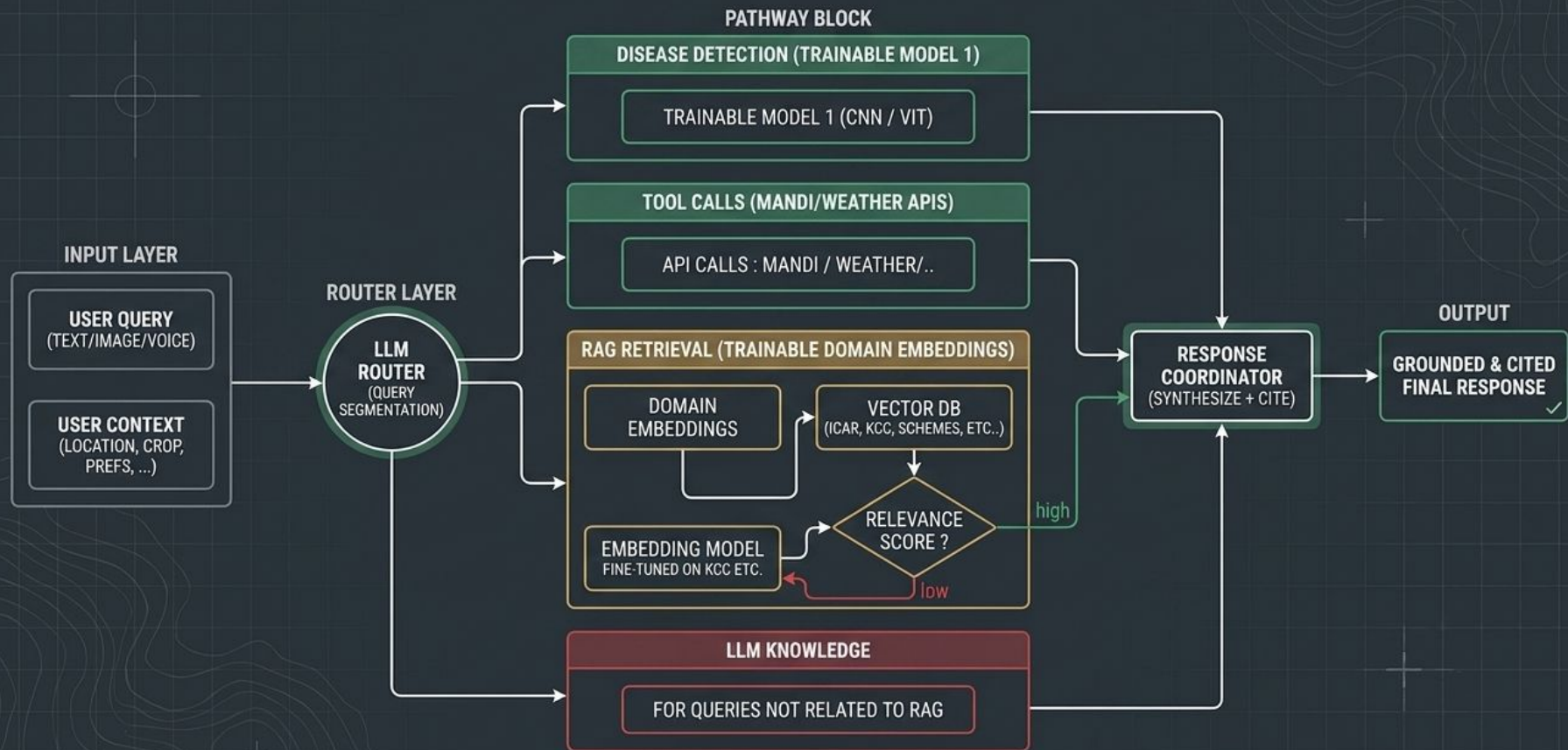


The Field (PlantDoc / Cassava)

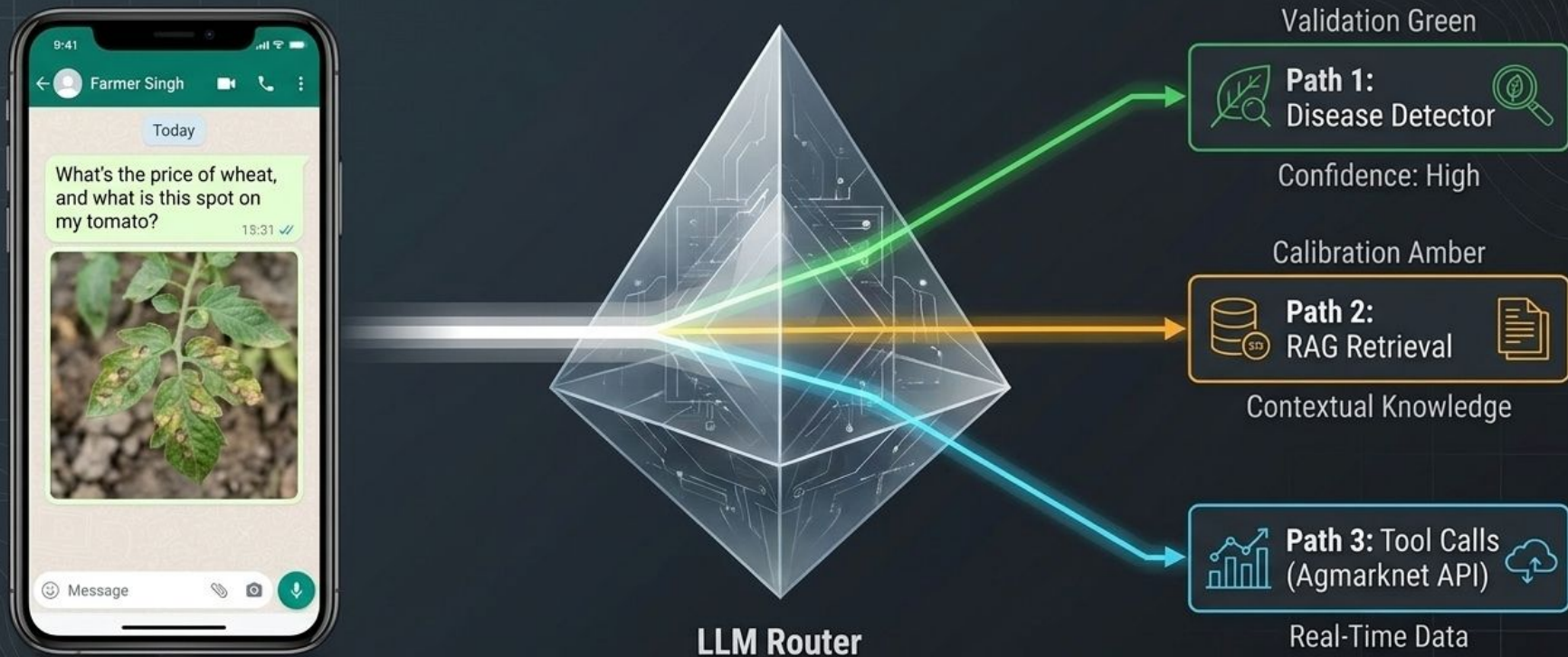


A farmer shooting a leaf in the sun against the soil is the failure case for standard AI. Most projects achieve 99% on PlantVillage and quietly stop. This system measures and narrows the gap.

MULTI-SOURCE AGENTIC RAG FRAMEWORK

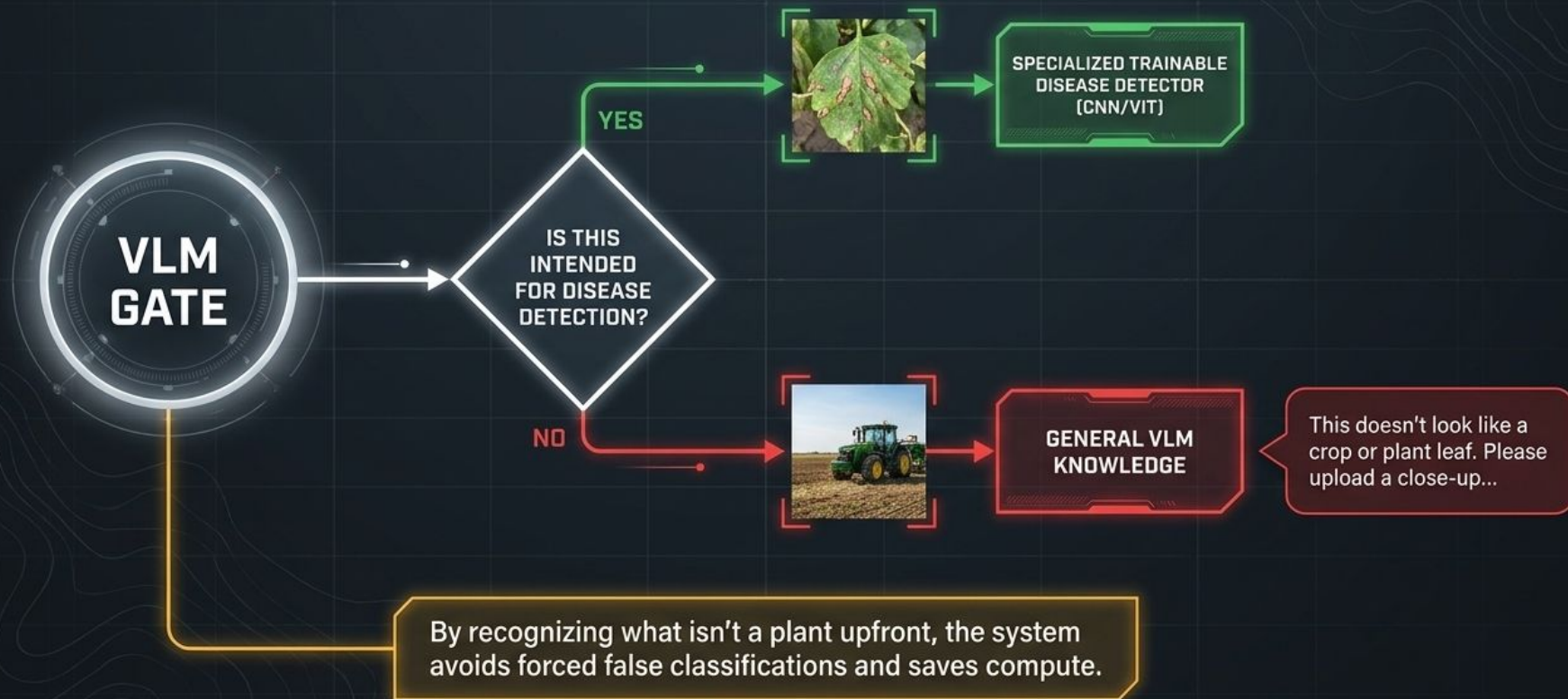


The LLM Router: Agentic Query Segmentation

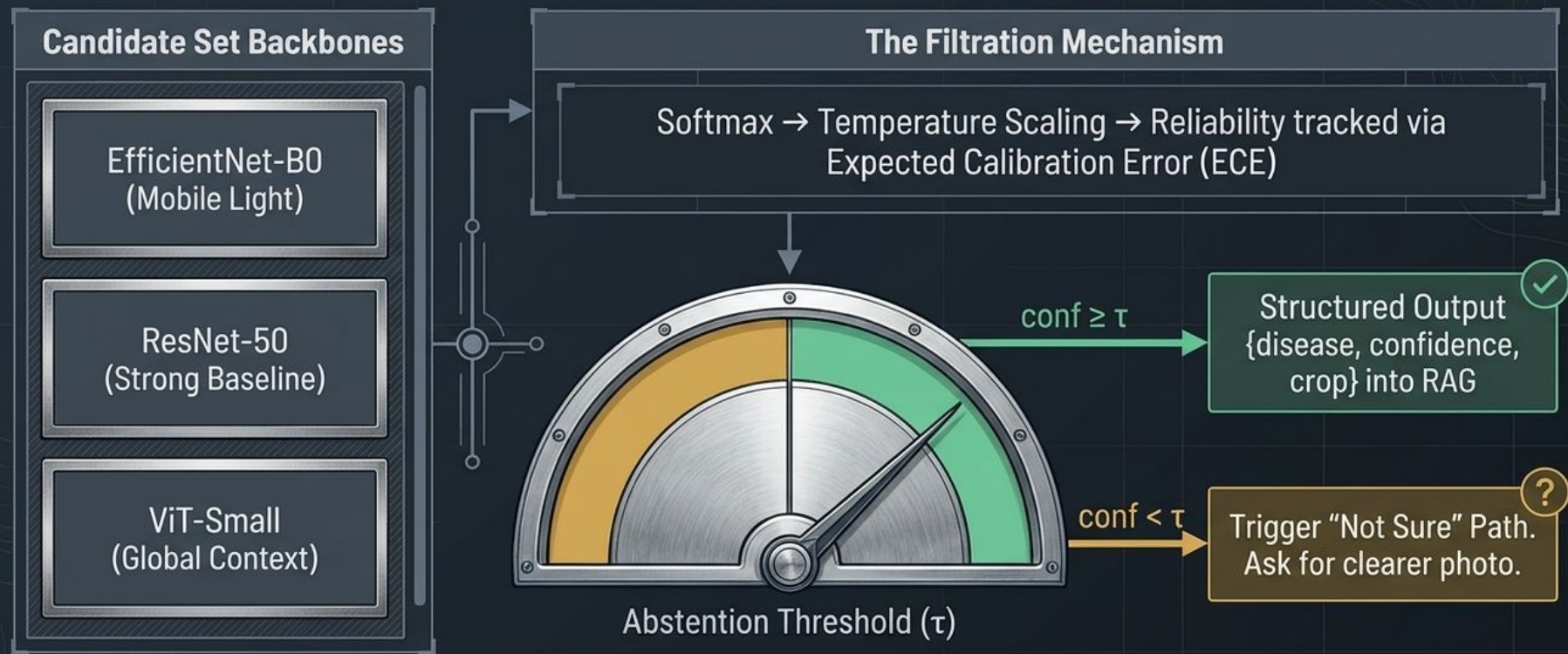


Insight: A single query can fire multiple paths simultaneously. The router prevents forcing multi-intent queries down a single bottleneck.

THE VLM PRE-GATE (FIRST FILTRATION)



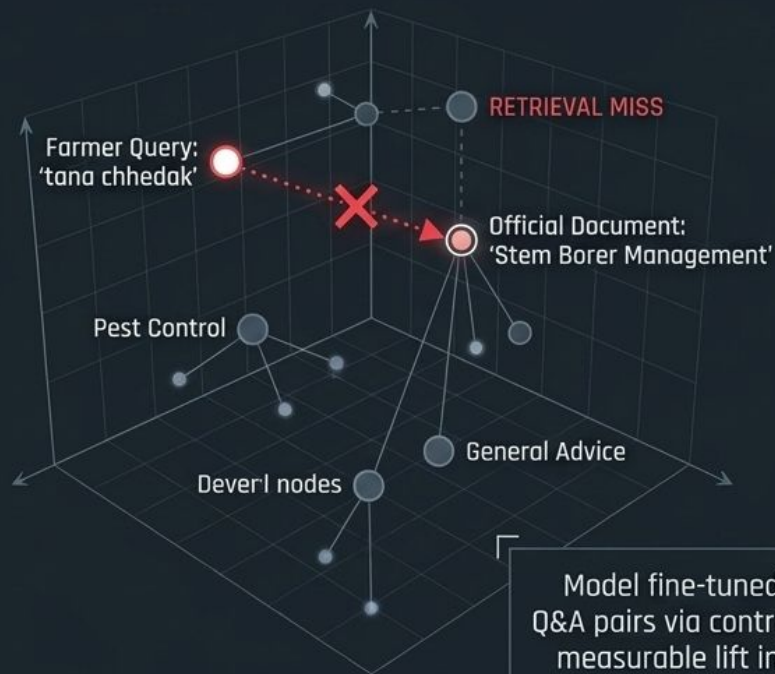
The Calibrated Disease Detector (Safety as a Mechanism)



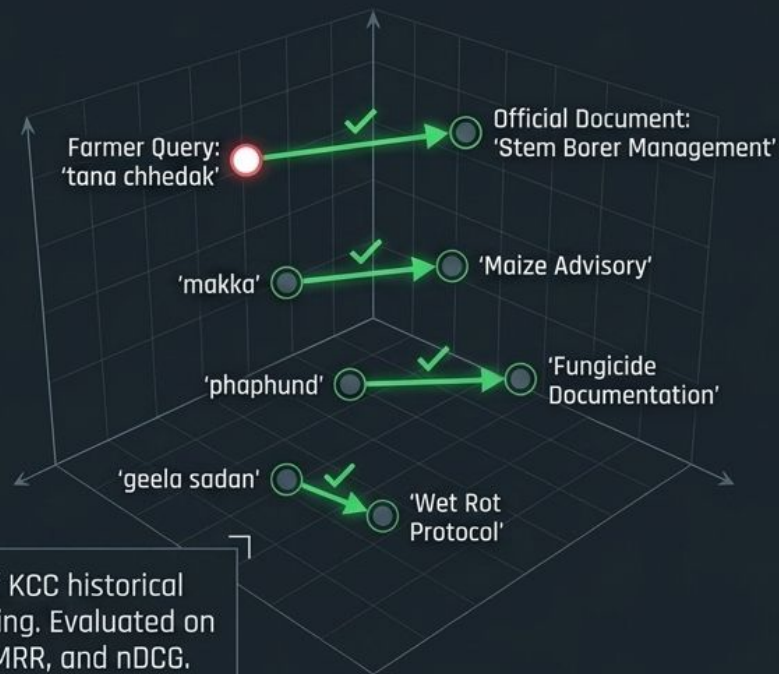
Safety is a measurable mathematical mechanism. The system gracefully asks for a clearer photo rather than guessing at 38%.

DOMAIN-ADAPTED EMBEDDINGS (CLOSING THE VERNACULAR GAP)

THE PROBLEM (GENERIC MODELS LIKE mBERT)



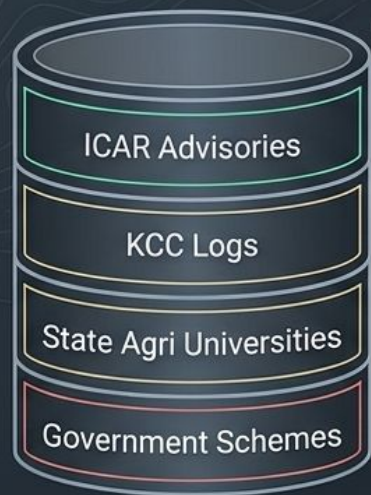
THE SOLUTION (FINE-TUNING VIA MuRIL/IndicBERT)



Model fine-tuned on lakhs of KCC historical Q&A pairs via contrastive learning. Evaluated on measurable lift in Recall@5, MRR, and nDCG.

🔍 Recall@5 ✓ ⬇️ MRR ✓ 📊 nDCG ✓

RAG Advisory Engine & The Tiered Response



**Vector DB
(ChromaDB)**

**Relevance
Score
Threshold**



Tier 1: Grounded (RAG Hit)

High relevance. Answer built ONLY from retrieved docs.



Tier 2: Model Knowledge (RAG Miss)

Low relevance. Valid farm query, but corpus is thin. Uses transparent fallback with clear LLM disclaimers.



Tier 3: Out of Scope

Non-agri query. Fails threshold. Politely redirects.



FAITHFULNESS IN ACTION: PROVING SYSTEM BOUNDARIES

Tier 1: Grounded (Score: 0.89)

How to control early blight in tomato?

Recommends Mancozeb
75% WP at 2.5 g/litre.
(Source: ICAR-IIVR,
2023)



• No hallucination, purely sourced.

Tier 2: Fallback (Score: 0.42)

Which wheat seed variety
for rabi season in Barmer?

⚠ Based on general
knowledge, not a verified
local advisory. Please
confirm with KVK as
availability changes.
General suggestion:
Raj 4120.

• Honest boundaries established.

Tier 3: Out of Scope

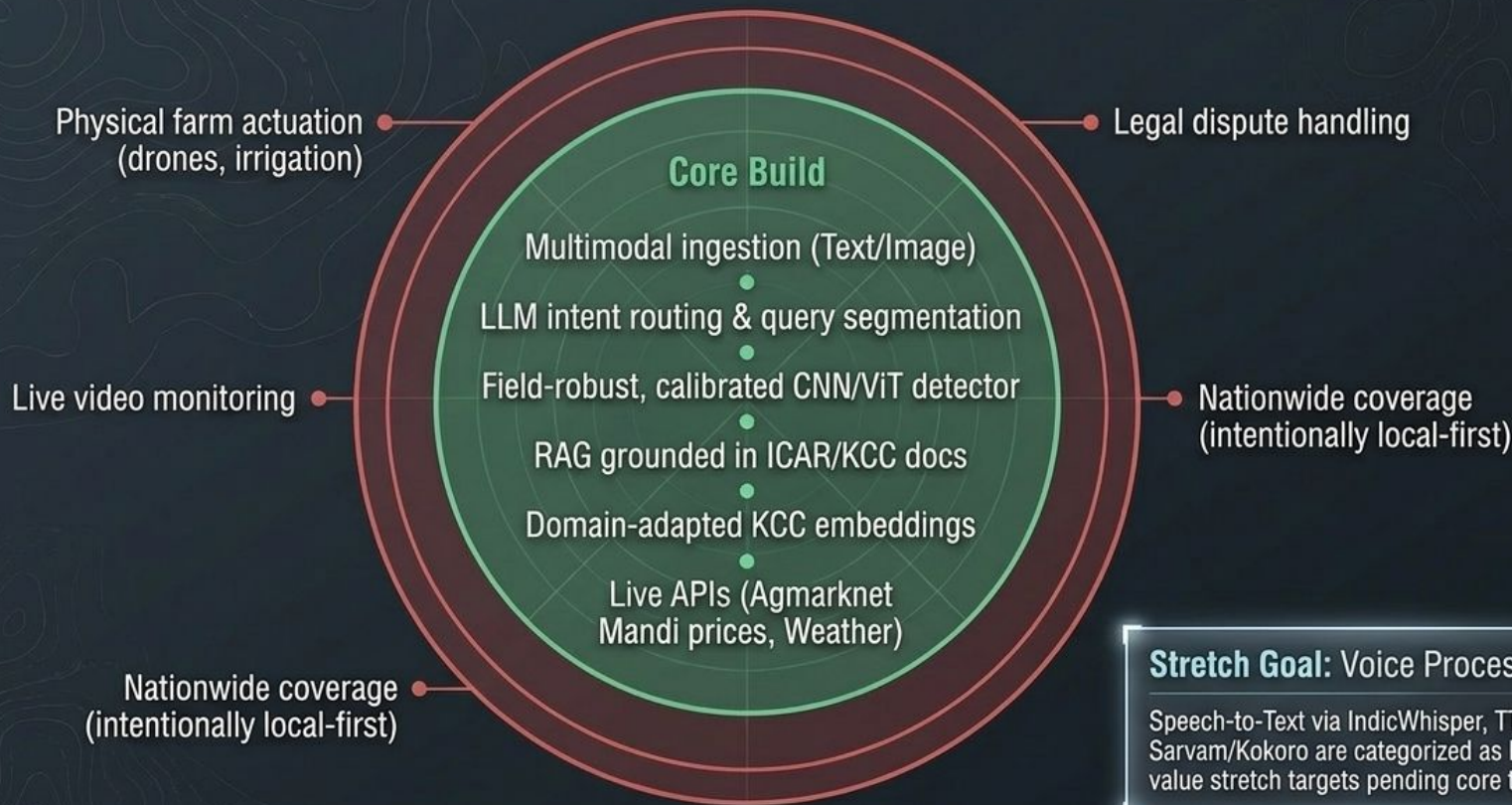
What's the cricket score?

I specialize in agricultural
advisories and cannot
provide cricket scores.
How can I help with your
farm today?



• Firm domain restriction.

Scope & Boundaries (Milestone 1 Execution)

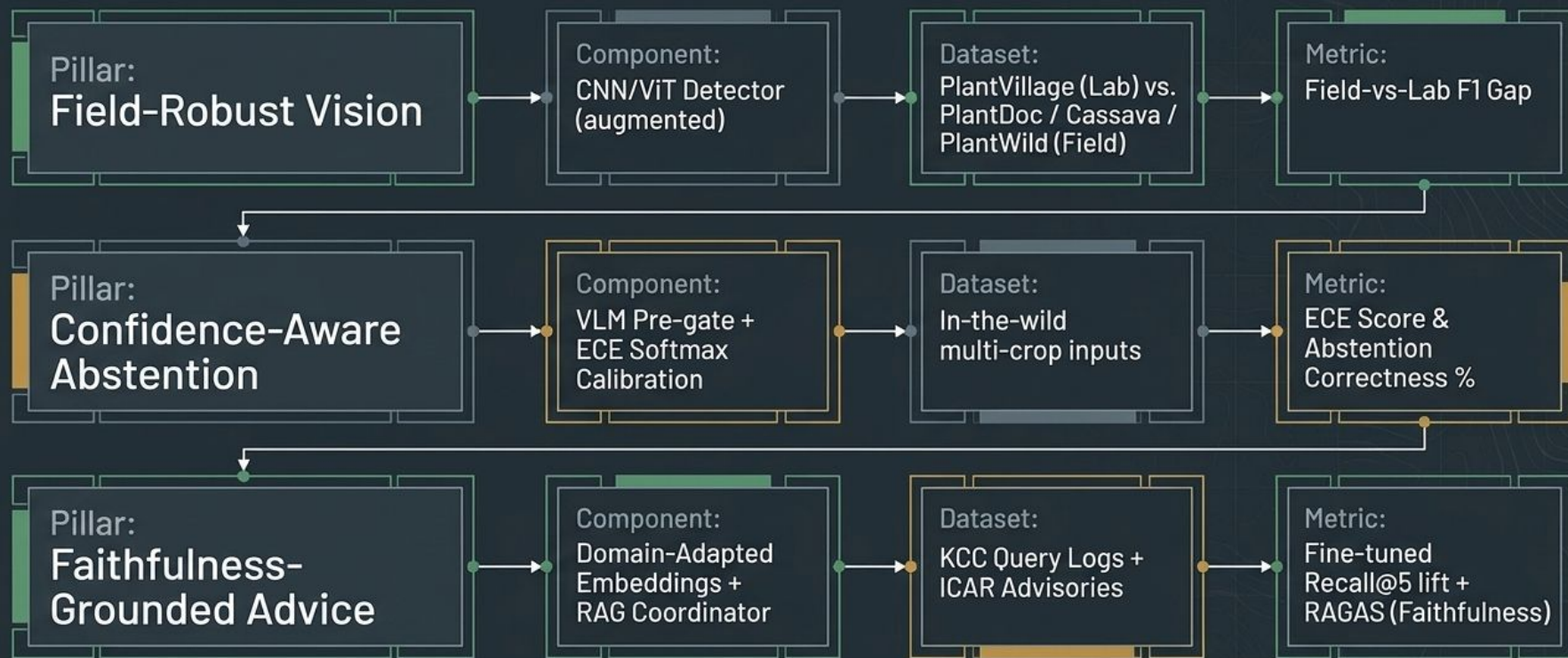


Comparative Landscape

Capabilities	KisanQRS	ReadyTensor	Krishi Sathi	RAG+Leaf	FarmerChat	Rooted in Truth (Ours)
LLM Generation	✗	✓	✓	✗	✓	✓
Image Detection	✓	✓	✗	✓	✗	✓
Multilingual/Hinglish	✓	✗	✓	✓	✓	✓
KCC-Data Backbone	✗	✗	✗	✗	✗	✓
Evaluated Field-Robustness	✗	✗	✗	✗	✗	✓
Confidence-Aware Abstention	✗	✗	✗	✗	✗	✓
Faithfulness Metrics (RAGAS)	✗	✗	✗	✗	✗	✓

Our system is the only one in the landscape that treats field-robustness, abstention, and faithfulness as continuously measured mathematical claims rather than assumed byproducts.

The Promises vs. Proof Alignment Matrix



Milestone 1 Fulfillment: A Verifiable Foundation

Achieved Metrics & Specifications

-  Problem statement & measurable objectives defined
-  System architecture designed & routed
-  Trainable model specs (Detector + Embeddings) selected
-  Public datasets identified (No scraping required)
-  Evaluation metrics (RAGAS, ECE, F1) locked

"Fluency is not accuracy.
By replacing blind
generative confidence
with **calibrated**
abstention and verifiable
retrieval, we bridge the
gap between **AI capability**
and **smallholder trust.**"